

Zafar Khan

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Professional Experience

CBOE Global Markets, Full Stack Software Engineer

December 2025 - Present

- Built a greenfield real-time operations dashboard for market data, trading ops, and analytics, processing 10M+ daily options events within a microservices architecture using Docker and Kubernetes (C#, React, TypeScript, PostgreSQL).
- Developed Jenkins CI/CD pipelines and integrated with existing distributed options alert services, delivering new microservices within two-week Agile sprints.
- Built a natural language database query via MCP, allowing non-technical production teams to query live PostgreSQL data without direct database access.
- Developed Claude Code skills for automated REST API best practice verification, integrated into the engineering workflow.
- Designed and implemented an RBAC system supporting 500 users with granular permission controls across the platform.

Wabtec Corporation, Full Stack Software Engineer

June 2022 - December 2025

- Designed and deployed a centralized license management system for external enterprise customers, reducing license provisioning from 30 minutes to under 5 minutes on \$100K+ deals (C#, Python, MySQL, AWS, Docker)
- Delivered a real time remote display system streaming live operational data for external enterprise customers, utilizing WebSockets and OpenSSL for bidirectional server-client communication containerized for deployment (C++, docker)
- Implemented an automated log monitoring pipeline using a machine learning model to detect known defects before customer reports, achieving 99% accuracy (Python, PyTorch)

Wabtec Corporation, Software Engineering Intern/Co-op

May 2021- May 2022

- Created a custom UI for engineers to simulate train runs (playback tool) and included functionality for accurate data to be read. (C# / XAML)
- Automated setup processes, optimizing log parsing and tool runtime efficiency to make the tool over 20x faster. (C#)

The University of Iowa, Undergraduate Teaching Assistant

January 2020 - May 2022

- Guided students in gaining an understanding of the basics of coding (C++)
- Graded homeworks, held office hours, proctored exams, and aided a class of over 300 students learning to code

Education

Penn State University

December 2024

Master of Science in Software Engineering

GPA: 3.8

The University of Iowa

May 2022

Bachelor of Science in Computer Science Engineering

GPA: 3.7 (Presidents List, Old Gold Scholar)

Activities/Projects

The University of Iowa SpaceX and Hyperloop Team, Software Lead

September 2018 - May 2022

- Worked in specific software group to develop and code certain aspects of hyperloop pod (C++)

National Voting System Application (Hackathon top 5 winner)

October 2020

- Developed a full-stack national voting app, featuring custom ballots, real-time voting, and result tracking. (Java / SQL)

Skills

C# • C++ • Python • TypeScript (HTML, CSS) • JavaScript • SQL • .NET • React • EF Core • Git
PostgreSQL • Jenkins • Agile • REST API Design • Kubernetes • PyTorch • LLM • Claude • Docker